Dockerized AI Application using Docker Compose

1. Objective

To build and deploy a **multi-container AI application** using Docker and Docker Compose consisting of:

- **Frontend** – Nginx serving HTML
- **Backend Model API** – Python Flask
- **Database** – MongoDB

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2. Prerequisites

Ensure the following are installed:

- Docker
- Docker Compose
- Python 3
- Git (optional)

Check installations:

```
docker --version
docker compose version
python3 –version
```

3. Project Folder Structure

Create the project directory.

```
mkdir ai-docker-app
cd ai-docker-app
```

Create the following structure:

```
ai-docker-app
│
├── docker-compose.yml
│
├── model
│   ├── app.py
│   ├── requirements.txt
│   └── Dockerfile
```

```
│
└── frontend
    ├── index.html
    └── Dockerfile
```

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4. Backend Model (Flask API)

Navigate to model folder.

```
mkdir model
cd model
```

---

app.py

```
from flask import Flask
from pymongo import MongoClient
app = Flask(__name__)
client = MongoClient("mongodb://mongodb:27017/")
db = client.testdb
@app.route("/")
def home():
    return "AI Model Running"
if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

requirements.txt

```
flask
pymongo
```

Model Dockerfile

Create a file named **Dockerfile**

```
FROM python:3.9
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY app.py .
CMD ["python", "app.py"]
```

5. Frontend Setup

Go back to project folder.

```
cd ..
mkdir frontend
cd frontend
```

---

index.html

```
<!DOCTYPE html>
<html>
<head>
<title>AI Docker App</title>
</head>
<body>
<h1>AI Docker Application</h1>
<p>Frontend running in Nginx container.</p>
</body>
</html>
```

Frontend Dockerfile

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

6. Docker Compose Configuration

Go back to project root.

```
cd ..
```

Create **docker-compose.yml**

```
version: "3"
services:
  mongodb:
    image: mongo
    container_name: mongo_db
    ports:
      - "27018:27017"
  model:
    build: ./model
    ports:
      - "5001:5000"
    depends_on:
      - mongodb

  frontend:
    build: ./frontend
    ports:
      - "8081:80"
    depends_on:
      - model
```

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7. Build and Run Containers

Run the following command from the project root:

```
sudo docker compose up –build
```

This command will:

1. Build model container
2. Build frontend container
3. Pull MongoDB image
4. Start all services

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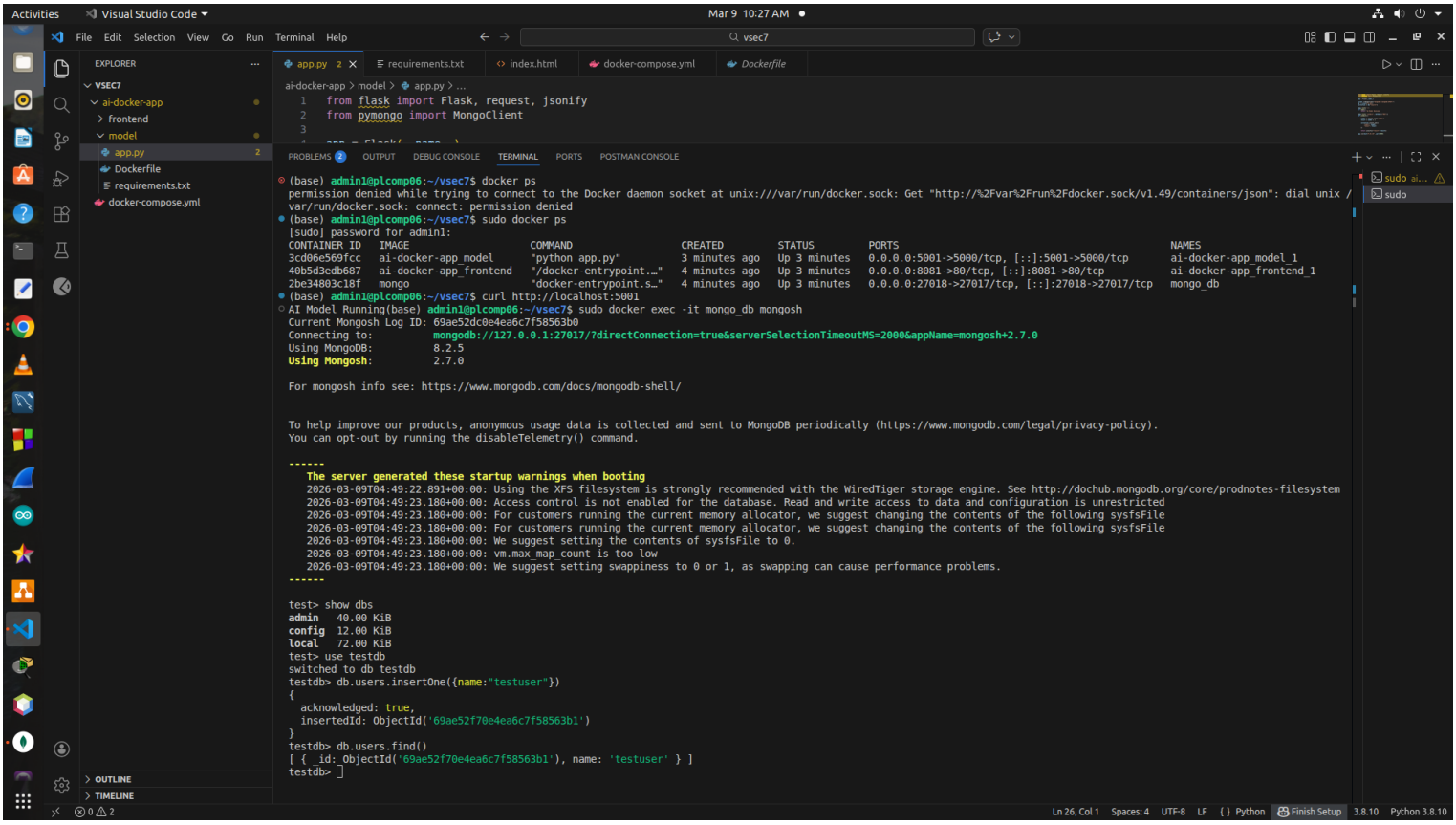
8. Verify Running Containers

Run:

```
sudo docker ps
```

Expected output:

```
ai-docker-app_model_1
ai-docker-app_frontend_1
mongo_db
```



9. Testing the Application

Test Frontend

Open browser:

http://localhost:8081

Expected output:

```
AI Docker Application
Frontend running in Nginx container.
```

Test Backend API

Run:

```
curl http://localhost:5001
```

Expected output:

```
AI Model Running
```

Test MongoDB

Access MongoDB container:

```
sudo docker exec -it mongo_db mongosh
```

Show databases:

```
show dbs
```

Create database:

```
use testdb
```

Insert data:

```
db.users.insertOne({name:"testuser"})
```

View data:

```
db.users.find()
```

Exit MongoDB:

```
Exit
```

---

10. Check Container Logs

Frontend logs:

```
sudo docker logs ai-docker-app_frontend_1
```

Model logs:

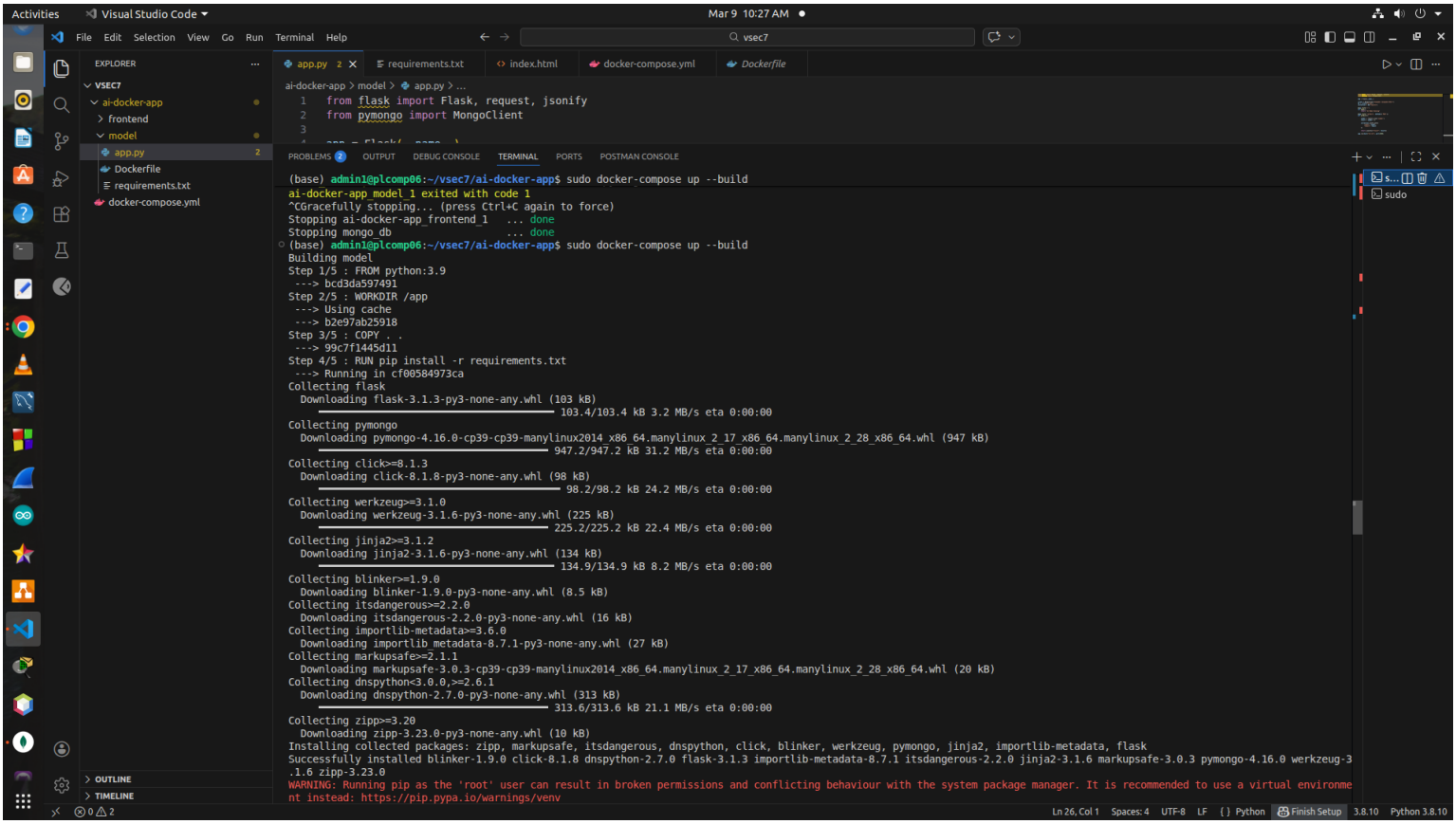
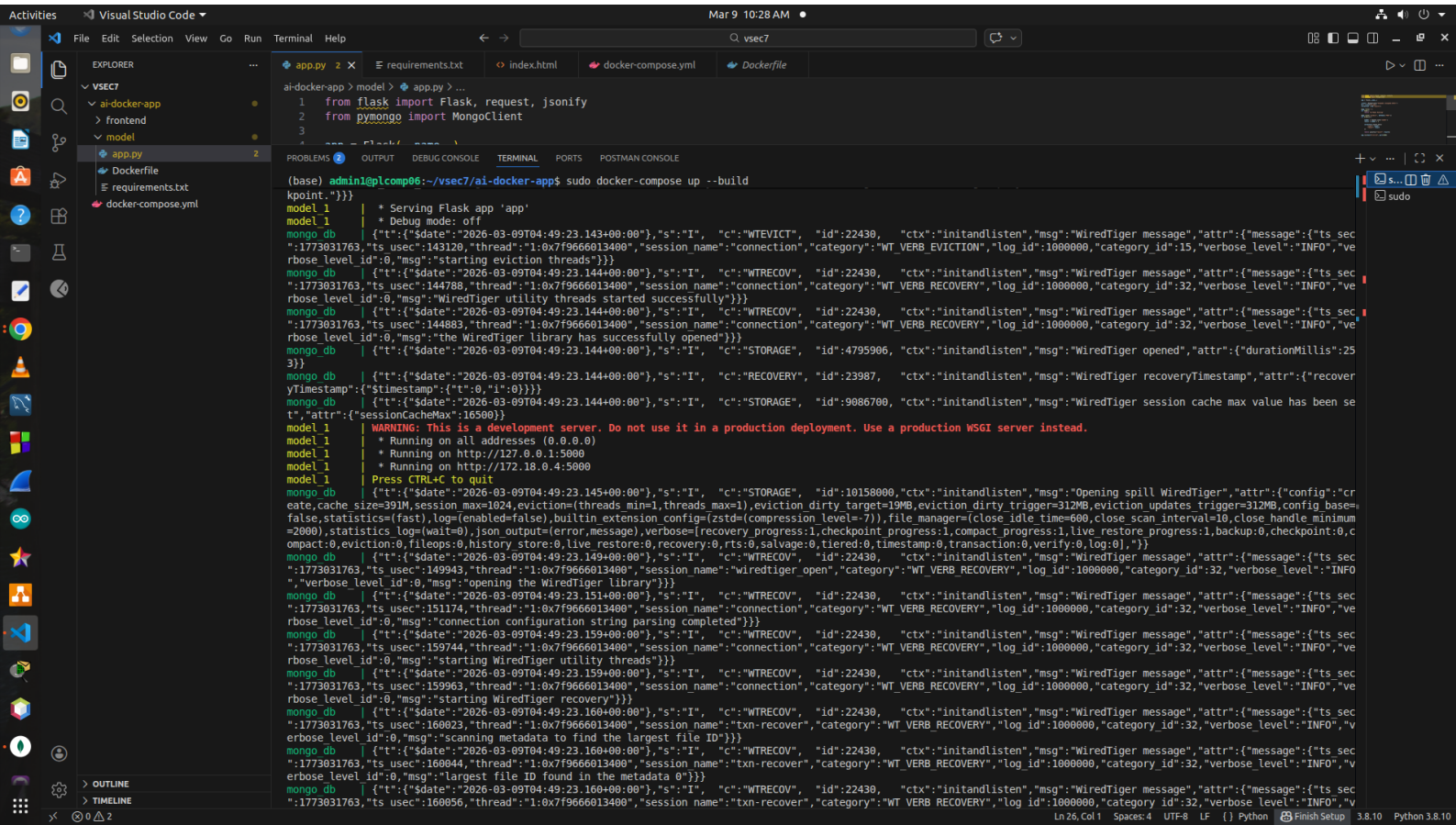
```
sudo docker logs ai-docker-app_model_1
```

MongoDB logs:

```
sudo docker logs mongo_db
```

---

11. Stopping Containers

To stop all services:

```
docker compose down
```